

Foundations of Bayesian network

[Bayesian network]

$$P(h \mid D) = \frac{P(D \mid h)P(h)}{P(D)}$$

1.0 Content Block

However, if the quantities are related, so that for example the individual θ_i have themselves been drawn from an underlying distribution, then this relationship destroys the independence and suggests a more complex model, e.g.,

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OpenBUGS – Open-source development of WinBUGS. SPSS Modeler – Commercial software that includes an implementation for Bayesian networks. Stan (software) – Stan is an open-source package for obtaining Bayesian inference using the No-U-Turn sampler (NUTS), a variant of Hamiltonian Monte Carlo. WinBUGS – One of the first computational implementations of MCMC samplers. No longer maintained.

3.0 Content Block

are equivalent: that is they impose exactly the same conditional independence requirements. A causal network is a Bayesian network with the requirement that the relationships be causal. The additional semantics of causal networks specify that if a node X is actively caused to be in a given state x (an action written as $\text{do}(X = x)$), then the probability density function changes to that of the network obtained by cutting the links from the parents of X to X , and setting X to the caused value x . Using these semantics, the impact of external interventions from data obtained prior to intervention can be predicted.

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Parameter learning In order to fully specify the Bayesian network and thus fully represent the joint probability distribution, it is necessary to specify for each node X the probability distribution for X conditional upon X 's parents. The distribution of X conditional upon its parents may have any form. It is common to work with discrete or Gaussian distributions since that simplifies calculations. Sometimes only constraints on distribution are known; one can then use the principle of maximum entropy to determine a single distribution, the one with the greatest entropy given the constraints. (Analogously, in the specific context of a dynamic Bayesian network, the conditional distribution for the hidden state's temporal evolution is commonly specified to maximize the entropy rate of the implied stochastic process.) Often these conditional distributions include parameters that are unknown and must be estimated from data, e.g., via the maximum likelihood approach. Direct maximization of the likelihood (or of the posterior probability) is often complex given unobserved variables. A classical approach to this problem is the expectation-maximization algorithm, which alternates computing expected values of the unobserved variables conditional on observed data, with maximizing the complete likelihood (or posterior) assuming that previously computed expected values are correct. Under mild regularity conditions, this process converges on maximum likelihood (or maximum posterior) values for parameters. A more fully Bayesian approach to parameters is to treat them as additional unobserved variables and to compute a full posterior distribution over all nodes conditional upon observed data, then to integrate out the parameters. This approach can be expensive and lead to large dimension models, making classical parameter-setting approaches more tractable.

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Marginal independence structure In general, learning a Bayesian network from data is known to be NP-hard. This is due in part to the combinatorial explosion of enumerating DAGs as the number of variables increases. Nevertheless, insights about an underlying Bayesian network can be learned from data in polynomial time by focusing on its marginal independence structure: while the conditional independence statements of a distribution modeled by a Bayesian network are encoded by a DAG (according to the factorization and Markov properties above), its marginal independence statements—the conditional independence statements in which the conditioning set is empty—are encoded by a simple undirected graph with special properties such as equal intersection and independence numbers.

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A Bayesian network (also known as a Bayes network, Bayes net, belief network, or decision network) is a probabilistic graphical model that represents a set of variables and their conditional dependencies via a directed acyclic graph (DAG). While it is one of several forms of causal notation, causal networks are special cases of Bayesian networks. Bayesian networks are ideal for taking an event that occurred and predicting the likelihood that any one of several possible known causes was the contributing factor. For example, a Bayesian network could represent the probabilistic relationships between diseases and symptoms. Given symptoms, the network can be used to compute the probabilities of the presence of various diseases. Efficient algorithms can perform inference and learning in Bayesian networks. Bayesian networks that model sequences of variables (e.g. speech signals or protein sequences) are called dynamic Bayesian networks. Generalizations of Bayesian networks that can represent and solve decision problems under uncertainty are called influence diagrams.

7.0 Content Block

The first 2 represent the same dependencies (X and Z are independent given Y) and are, therefore, indistinguishable. The collider, however, can be uniquely identified, since X and Z are marginally independent and all other pairs are dependent. Thus, while the skeletons (the graphs stripped of arrows) of these three triplets are identical, the directionality of the arrows is partially identifiable. The same distinction applies when X and Z have common parents, except that one must first condition on those parents. Algorithms have been developed to systematically determine the skeleton of the underlying graph and, then, orient all arrows whose directionality is dictated by the conditional independences observed. An alternative method of structural learning uses optimization-based search. It requires a scoring function and a search strategy. A common scoring function is posterior probability of the structure given the training data, like the BIC or the BDeu. The time requirement of an exhaustive search returning a structure that maximizes the score is superexponential in the number of variables. A local search strategy makes incremental changes aimed at improving the score of the structure. A global search algorithm like Markov chain Monte Carlo (MCMC) can avoid getting trapped in local minima. Finding a structure that maximizes the mutual information, typically by restricting the parent candidate set to k nodes or by finding an optimal k on a node-per-node basis, is a technique that consistently achieves high scores on benchmark datasets. A particularly fast method for exact BN learning is to cast the problem as an optimization problem, and solve it using integer programming. Acyclicity constraints are added to the integer program (IP) during solving in the form of cutting planes. Such method can handle problems with up to 100 variables. In order to deal with problems with thousands of variables, a different approach is necessary. One is to first sample one ordering, and then find the optimal BN structure with respect to that ordering. This implies working on the search space of the possible orderings, which is convenient as it is smaller than the space of network structures. Multiple orderings are then sampled and evaluated. This method has been proven to be the best available in literature when the number of variables is huge. Another method consists of focusing on the sub-class of decomposable models, for which the MLE have a closed form. It is then possible to discover a consistent structure for hundreds of variables. Learning

Bayesian networks with bounded treewidth is necessary to allow exact, tractable inference, since the worst-case inference complexity is exponential in the treewidth k (under the exponential time hypothesis). Yet, as a global property of the graph, it considerably increases the difficulty of the learning process. In this context it is possible to use K-tree for effective learning.